



The Maths of Predators and Preys

Don't get Eaten by the Wolf

Hares (No Predators)



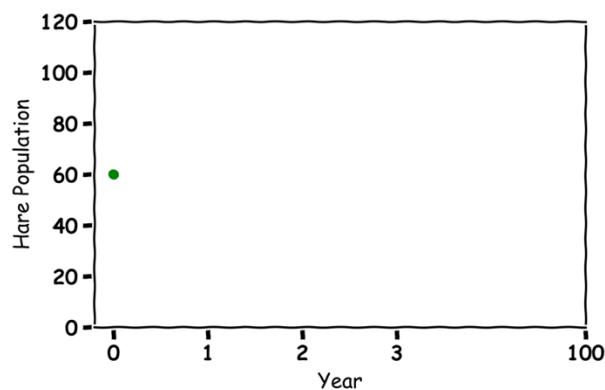
Given a population of 60 Hares and a **birth rate of 0.2** what would the population be after three years?

$$H_{\text{Future}} = H + 0.2(H)$$

Year	Hare Population
0	60
1	
2	
3	

Sketch the Graph.

Guess what would happen in 100 years?



Lynx (No Prey)



Given a population of 30 Lynxes and a **death rate of 0.1** what would the population be after three years?

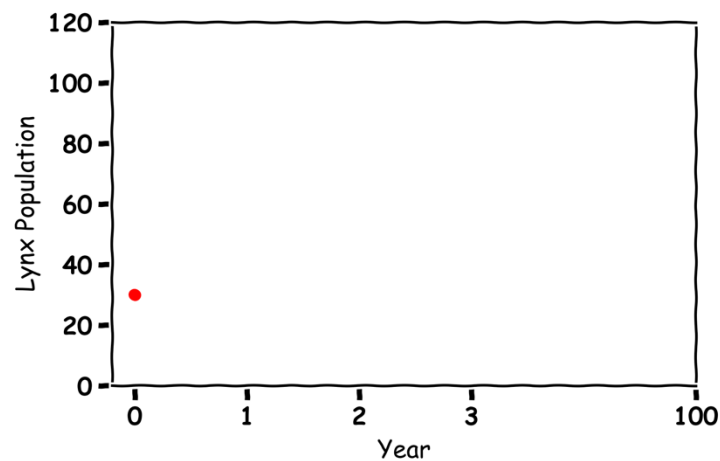
$$L_{\text{Future}} = L - 0.1(L)$$

Year	Lynx Population
0	30
1	
2	
3	



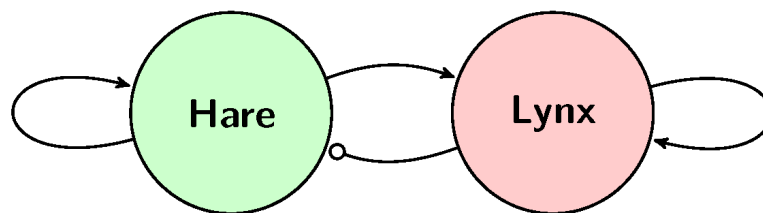
Sketch the Graph.

Guess what would happen in 100 years?



Hares and Lynxes

Label the graph:



Given a population of **60 Hares** and **30 Lynxes** in year 0 with a **birth rate of 0.2**, a **death rate of 0.1**, a **eat rate of 0.005** and a **food rate of 0.002** what would the populations be after three years?

Hare	Value	Lynx	Value
Birth	0.2	Death	0.1
Eat	0.005	Food	0.002

$$H_{\text{Future}} = H + 0.2(H) - 0.005(H)(L)$$

$$L_{\text{Future}} = L - 0.1(L) + 0.002(L)(H)$$

Year	Hare Population	Lynx Population
0	60	30
1		
2		
3		